Hints: Integrals – 4. Series and integrals

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Write $x^{-x} = e^{-x\ln x} = \sum_{n\geqslant 0} \frac{(-x\ln x)^n}{n!}$ and $x^x = \sum_{n\geqslant 0} \frac{(x\ln x)^n}{n!}$. You need $\int_0^1 x^n \ln^n x\,dx$.

Hint 2. Substituting $x = e^{-u}$: $\int_0^1 x^n \ln^n x\,dx = (-1)^n \int_0^\infty u^n e^{-(n+1)u} du = \frac{(-1)^n n!}{(n+1)^{n+1}}$. Plugging in gives $\sum_{n\geqslant 0}(n+1)^{-(n+1)}$ and $\sum_{n\geqslant 0}(-1)^n(n+1)^{-(n+1)}$.

Hint 3. Justification: for x^{-x} all terms are $\geqslant 0$ on $(0,1]$ (since $-x\ln x \geqslant 0$), so monotone convergence applies; for x^x the series is dominated by $e^{|x\ln x|} \leqslant e^{1/e}$. Numerically $\int_0^1 x^{-x}dx = 1 + \frac{1}{4} + \frac{1}{27} + \frac{1}{256} + \cdots \approx 1.291$.

Problem 2

Hint 1. For $x > 0$, $\frac{1}{e^x - 1} = \sum_{k\geqslant 1} e^{-kx}$, and $\int_0^\infty x^{s-1}e^{-kx}dx = \frac{\Gamma(s)}{k^s}$ (substitute $u = kx$).

Hint 2. For real $s > 1$ the terms are nonnegative, so summation and integration commute (monotone convergence); for complex s with $\operatorname{Re} s = \sigma > 1$ dominate by the case $s = \sigma$. Then $s = 2$: $\Gamma(2)\zeta(2) = \frac{\pi^2}{6}$; $s = 4$: $\Gamma(4)\zeta(4) = 6 \cdot \frac{\pi^4}{90} = \frac{\pi^4}{15}$.

Problem 3

Hint 1. All four are geometric series or derivatives of one: (a) $\sum_n \binom{n+3}{3} x^n = \frac{1}{(1-x)^4}$; (b) $\sum_{k\geqslant 1}(-1)^k x^{2k} = -\frac{x^2}{1+x^2}$; (c) $\sin\frac{k\pi}{2}$ runs through $0, 1, 0, -1, \ldots$, so the sum is $x - x^3 + x^5 - \cdots = \frac{x}{1+x^2}$; (d) $\frac{x^2}{1-x}$.

Hint 2. Answers: (a) $\left[\frac{1}{3(1-x)^3}\right]_0^{1/2} = \frac{7}{3}$; (b) $\frac{\pi}{4} - 1$; (c) $\frac{1}{2}\ln(1+x^2) + C$ (for $|x| < 1$, where the series converges); (d) $\ln 2 - \frac{5}{8}$. In (b) the series diverges at $x = 1$, but it converges on $[0,1)$ and the closed form is integrable, so the integral is understood as the integral of the sum there.

Problem 4

Hint 1. Multiply out $(1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^m})$: by uniqueness of binary expansions every power x^j with $0 \leqslant j < 2^{m+1}$ appears exactly once.

Hint 2. So $\prod_{k\geqslant 0}(1+x^{2^k}) = \frac{1}{1-x}$ for $0 \leqslant x < 1$ (equivalently, the finite product equals $\frac{1-x^{2^{m+1}}}{1-x}$). The integrand is $1 - x$ and the integral is $\frac{1}{2}$.

Problem 5

Hint 1. Expand $\ln(1-x) = -\sum_{n\geqslant 1}\frac{x^n}{n}$ resp. $\frac{1}{1-x} = \sum_{n\geqslant 0} x^n$; all terms have constant sign on $(0,1)$, so termwise integration is justified. Use $\int_0^1 x^n \ln x\,dx = -\frac{1}{(n+1)^2}$ and $\int_0^1 x^n \ln^2 x\,dx = \frac{2}{(n+1)^3}$.

Hint 2. First: $\sum_{n\geqslant 1}\frac{1}{n(n+1)^2} = \sum\left(\frac{1}{n} - \frac{1}{n+1} - \frac{1}{(n+1)^2}\right) = 1 - (\zeta(2) - 1) = 2 - \frac{\pi^2}{6}$. Second: $2\zeta(3)$.